

Entity Linking over Historical Documents

Linking Casellario Politico Centrale records to Wikidata entities

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1. Motivation

Historical archives contain vast amounts of biographical information locked in semi-structured records that are difficult to query, cross-reference, or integrate with modern knowledge infrastructure. Connecting these records to public knowledge graphs such as Wikidata unlocks their scholarly and computational value: it enables disambiguation of individuals, enrichment with additional biographical facts, and interoperability with other digital-humanities resources.

This project addresses entity linking over the **Casellario Politico Centrale (CPC)**, a police-surveillance archive of the Italian state held by the Archivio Centrale dello Stato. The database contains 152,589 records, each referring to a person of political interest. The goal is to link these person mentions to the corresponding entities in Wikidata, exploiting the contextual information available in each record — names, dates of birth, places, occupations and political affiliation.

The central difficulty is that the linking decision must be made from sparse, noisy and multilingual archival fields against a knowledge base whose coverage of these individuals is very low. Most CPC persons simply have no Wikidata entry, and among those that do, superficial name matches are frequently spurious. A reliable pipeline must therefore combine automated candidate retrieval with a discriminative model that favours precision, since incorrectly linking two distinct people is more damaging than leaving a record unlinked.

Following the project guidelines, every design decision is documented and justified for explainability, and open, reproducible tools are preferred throughout.

2. Related Work

Entity linking (EL) is the task of associating a textual mention with the correct entry in a reference knowledge base, and typically decomposes into candidate generation and candidate ranking / disambiguation. Applying EL to digitised historical corpora is notably harder than the newswire setting: records are noisy, multilingual, sparsely attributed, and dominated by long-tail entities with little or no knowledge-base coverage.

Recent work situates this project within the historical-EL literature. Graciotti et al. study historical entity linking in the cultural-heritage domain in “*Musical Heritage Historical Entity Linking*” (Artificial Intelligence Review, 2025), addressing the disambiguation of historical mentions against knowledge graphs. The same group introduces *KE-MHISTO* (Findings of ACL, 2025), a multilingual historical knowledge-extraction benchmark explicitly targeting the long-tail problem — precisely the regime in which the CPC operates, where the overwhelming majority of persons are absent from the knowledge base.

Two families of approaches frame our own choices. Knowledge-base-driven retrieval methods query the target graph directly by name (here, Wikidata's entity search and SPARQL endpoints) to generate candidates; and feature-based disambiguation methods score each candidate using structured attribute agreement — dates, places, occupations — rather than relying on surface string similarity alone. LLM-based and hybrid methods are an increasingly common alternative for the ranking step. In this work we deliberately adopt a transparent, feature-based discriminative approach so that every linking decision remains inspectable and explainable, and we use it as the basis for comparing several classifiers of increasing sophistication against a simple heuristic baseline.

3. Methodology

The methodology is organised as a pipeline that turns raw archival records into a labelled dataset and, from it, an automatic linking model. It has four conceptual stages: (i) data acquisition and field selection, (ii) candidate generation and attribute enrichment from Wikidata, (iii) the design of comparison metrics between the two sources, and (iv) model design and a held-out evaluation protocol.

3.1 Data acquisition and field selection

The public CPC web interface is designed for human consultation and is unsuitable for bulk retrieval. By inspecting the network calls issued during manual queries, we identified the underlying Sour endpoint and its parameters (query restrictions, start row, number of rows and JSON output format), which allowed the complete database of 152,589 records to be downloaded automatically.

Each record consists exclusively of text fields. After analysing null-value rates and the information actually available in Wikidata, fields were pruned to those with discriminative power. Redundant fields (duplicated denominations and combined date/geography fields) and archival-reference fields (folder, file, registration numbers and internal annotations) were discarded, as they carry no information that helps identify the person. Among the location fields, only country and municipality of birth were retained, since these are the geographic attributes typically recorded in Wikidata; residence location was dropped after contrast tests showed it lacked discriminative power, being tied to the moment of registration rather than to the person. Registration start/end dates and the mostly-empty image references were likewise excluded.

Category	Field	Description	Nulls
Indexing	id	Unique key	0%
	id_text	Same as id field	0%
Personal identification	DENOMINAZIONE	COGNOME + NOME	0%
	NOME	Given name	0.43%
	COGNOME	Surname	0%
	PATERNITA	Father name	3.60%
	ALIAS	Alternative name	99.18%
Archive localization	BUSTA		0%
	FASCICOLO		25,58%
	MATRICOLA		83,19%
	NOTA_ARCHIVISTICA		99,79%
	NOTEDOCARC		77,06%
	ANNOTAZIONI		22,57%
Personal data	SESSO	Gender of person	0%
	COLORE	Political orientation	3,38%
	MESTIERE	Occupation	0,98%
Place of birth	NASCITA		3,22%
	CONTINENTE_DI_NASCITA		3,26%
	GEO_NASCITA		3,26%
	NAZIONE_DI_NASCITA		3,26%
	REGIONE_DI_NASCITA		3,26%
	PROVINCIA_DI_NASCITA		3,26%
	COMUNE_DI_NASCITA		3,22%
	RESIDENZA		2,63%

Place of residence	CONTINENTE_DI_RESIDENZA		2,65%
	GEO_RESIDENZA		2,65%
	NAZIONE_DI_RESIDENZA		2,65%
	REGIONE_DI_RESIDENZA		2,65%
	PROVINCIA_DI_RESIDENZA		2,62%
	COMUNE_DI_RESIDENZA		2,62%
Fechas	DATA_NASCITA	Birth year	4,62%
	DATAINIZIO	Initial year of registration	0,09%
	DATAFINE	Final year of registration	0,09%
	UNITDATE	DATAINIZIO + DATAFINE	0,07%
Referencias	FOTO	File reference	99,50%
	IMMAGINE_FASCICOLO	File reference	77,79%

Tab. 1 Fields of CPC database

The fields retained as identifying evidence were: first name, surname, father's name, year of birth, sex, occupation, political colour (affiliation), and country and municipality of birth.

3.2 Sampling for annotation

Because the whole database cannot be verified manually, representative subsets were drawn at random for labelling and for testing. An incremental, probability-based sampling function was implemented so that the subsequent web-querying stages — which may suffer service interruptions — can resume without loss. A labelling subset of 6,042 records and a disjoint test subset of 3,061 records were selected.

3.3 Candidate generation via Wikidata search

Candidates were generated using Wikidata's entity search service, which returns candidate QIDs for a text string. Three deliberate choices govern this step:

- **Query construction.** Wikidata expects person queries as first name followed by surname; the NOME and COGNOME fields were used accordingly. When a record contains several given names, both the full string and each given name paired with the surname were queried, since this occasionally surfaced valuable matches.
- **Query language.** In all tests, results in Italian ('it') were identical to those in English ('en'), so only Italian queries were issued.
- **Result cap.** A maximum of 20 results per query was set: cases exceeding 20 corresponded to records missing the given name where the surname is a common polysemous word, for which a reliable match is very unlikely in any case.

Each candidate is stored with its query string, name variant flag, Wikidata label, description and aliases. After this stage a person may have several candidate rows, or none if the search returned nothing.

3.4 Attribute enrichment via SPARQL

For every retrieved QID, SPARQL queries against Wikidata fetched the properties needed to compare against the CPC record, establishing the following correspondences:

CPC field	Wikidata property	Description
—	wdt:P31	Instance / entity type
NOME	wdt:P735	Given name
COGNOME	wdt:P734	Family name

CPC field	Wikidata property	Description
PATERNITA	wdt:P22/wdt:P735	Father's name
DATA_NASCITA	wdt:P569	Date of birth
SESSO	wdt:P21	Sex
MESTIERE	wdt:P106	Occupation
NAZIONE_DI_NASCITA	wdt:P27	Country of birth
COMUNE_DI_NASCITA	wdt:P19	Place of birth

Tab. 2 Entity properties association

The candidate aliases and the number of entries returned by the search were also attached to each pair, on the intuition that a person yielding many candidate entries is a less reliable match whose evidence requirements should be stricter.

3.5 Comparison metrics

With paired fields from both sources, each attribute is reduced to a numeric agreement metric. After extensive exploration, most metrics take one of three values: +1 when the values are judged to correspond, 0 when a value is missing, and -1 when they do not correspond or correspondence is impossible to establish. A two-valued alternative (treating a missing field the same as a mismatch) was tested but slightly degraded model performance, so the three-valued scheme was kept.

Metric-specific decisions:

- **Year of birth** (one of the most relevant attributes): +1 for an exact match, 0 for a one-year difference, and -1 for larger differences.
- **City of birth**: the only field using a Levenshtein-based string similarity, producing a real value in [0, 1], because small spelling variants and supplementary text genuinely occur here.
- **Country of birth**: the value space is small, so detected variants were encoded directly (e.g. Yugoslavia / Jugoslavia, Repubblica di San Marino / San Marino, Italia / Regno d'Italia).
- **Political affiliation** (COLORE): the heuristic checks whether the CPC term appears (+1) or not (0) in the Wikidata description.
- **Occupation**: treated as sets, checking for a non-empty intersection. Because values often add extra detail and use synonyms, a combination of regular-expression filters and a small 8-entry synonym dictionary was used; Levenshtein distance was discarded here as it increased false positives without adding value.

This yields eight comparison features per candidate pair: eval_birth, eval_name, eval_sex, eval_country, eval_city, eval_father, eval_occupation and eval_affiliation.

3.6 Manual labelling protocol

A gold label was assigned to each remaining candidate pair through manual verification that went beyond the field content, including online research to confirm the reliability of each assignment. Precision was prioritised: in doubtful cases — typically caused by missing data — the pair was always labelled as not linked.

3.7 Models and evaluation protocol

An inspection of more than 1,000 records found no correct correspondence with a birth-year difference greater than one year. This justifies classifying all pairs with a difference above one year as negative, which transforms an extremely imbalanced problem (78 positives against 4,779 negatives) into a balanced one (78 against 54) where standard classifiers can be applied and standard metrics (precision, recall, F1) are meaningful.

Four models of increasing sophistication were selected; two further options were considered and rejected:

Model	Included	Rationale
Heuristic rule	Yes	Problem analysis and data inspection show that a few simple rules can discriminate a significant number of cases.
Logistic Regression	Yes	With eight bounded numeric features, a classic linear method is a natural check for adequate class separation.
Random Forest	Yes	Tests whether learned decision trees behave better than the single heuristic condition.
Gradient Boosting (XGBoost-style)	Yes	Gradient-boosted tree ensembles expose the best decision logic the trees can find.
Simple neural network	No	Only 132 labelled cases; a network would memorise the table or collapse to the simpler methods. Reaching ~2,000 cases would require labelling 70% of the data.
Tabular NN + LLM	No	Feeding raw textual fields to a language-capable network exceeds the available technical and hardware resources.

Tab. 3 Models taken into consideration

Because several decisions — in particular the heuristic rule and its threshold — were taken after examining the labelling set, a disjoint test set was used to confirm that the models generalise.

4. Implementation

The pipeline is implemented in three python modules: `labellingprocess.py`, which builds the labelled dataset from raw records, and `evaluation.py`, which trains and evaluates the models. Each stage is a documented function that reads the output of the previous stage and writes an intermediate file, so the whole process is incremental and reproducible.

4.1 Dataset-construction stages

Function	Output file	Records	Purpose
download_cpc	cpc_records.jsonl	152,589	Full CPC database via the Solr endpoint
add_subset	cpc_extract.jsonl	6,042	Random labelling subset (incremental)
step1b	step1.csv	11,369	Wikidata search candidates per person
step2a	step2a.csv	5,465	SPARQL attribute enrichment of candidates
step2b	step2b.csv	5,465	Adds aliases and candidate-count features
step3	step3.csv / labelling.csv	5,465 / 132	Comparison metrics; filtered gold-labelling set

Tab. 4 Main functions in dataset construction

The candidate-generation stage (step1b) produced 11,369 candidate rows for the 6,042 labelling persons: **4,611 persons returned no results** and were assigned to the not-linked category, while the remaining 1,431 persons produced 3.82 Wikidata candidates each on average, giving 5,465 (CPC, Wikidata) pairs carried forward. The step3 function emits two files: the full set with all metrics, and `labelling.csv`, restricted to pairs whose birth-year metric is 0 or +1 and augmented with an empty *gold* column for annotation.

4.2 Model implementations

The four models are implemented in Evaluation.ipynb:

- **Heuristic** (heuristic(X, y, threshold)): predicts Yes when eval_occupation = 1 or eval_city > threshold, a rule derived from the two most discriminative features.
- **Logistic Regression** (regression(X, y)): scikit-learn LogisticRegression; the largest positive coefficients are eval_city (+2.04), eval_country (+1.37) and eval_occupation (+1.33).
- **Random Forest** (forest(X, y)): scikit-learn RandomForestClassifier with 300 trees and depth capped at 3 to prevent overfitting on 132 rows, with class-imbalance weighting enabled. Feature importance is dominated by eval_city (0.43), eval_birth (0.22) and eval_occupation (0.16).
- **Gradient Boosting** (xgboost(X, y)): scikit-learn GradientBoostingClassifier with 100 estimators, depth 3 and imbalance weighting. Its most important tree relies on the same features as the heuristic (eval_city, eval_occupation) but with a slightly more sophisticated logic; feature importance is concentrated on eval_city (0.80) and eval_occupation (0.15).

5. Evaluation

Two datasets are used. The labelling set was built as described above; the disjoint test set was constructed by the same pipeline over 3,061 different, randomly chosen persons. In the test set, 2,330 persons returned no candidate, and after filtering non-person Wikidata entries and pairs whose birth year differed by more than one year, 75 pairs (74 distinct persons) remained — 55 positive (73.3%) and 20 negative (26.7%). The classes are somewhat more imbalanced than in the labelling set, but not enough to invalidate the evaluation.

Labelling set. After candidate retrieval and enrichment, 5,465 pairs remained; removing non-person entries left 4,857, and removing pairs with a birth-year gap above one year left 132 pairs (122 persons). Of these, **78 pairs (59.1%) were labelled positive** — 76 distinct persons — and 54 (40.9%) negative, giving a well-balanced dataset.

Heuristic threshold. Precision and recall were traced across candidate thresholds for eval_city. From a value of **0.85** upward the heuristic produced no false positives. Since precision is prioritised over recall — an incorrect link being more harmful than a missed one — the threshold was fixed at 0.85 for all subsequent evaluation.

All four models were then evaluated on both sets with accuracy, precision, recall and F1, using the confusion-matrix counts recorded during the experiments.

6. Results

The table below summarises performance across all models on both datasets.

Model / set	Accuracy	Precision	Recall	F1
Heuristic — labelling	96.2%	100.0%	93.6%	0.967
Heuristic — test	97.3%	100.0%	96.4%	0.981
Logistic Reg. — labelling	93.2%	92.6%	96.2%	0.943
Logistic Reg. — test	96.0%	100.0%	94.5%	0.972
Random Forest — labelling	96.2%	100.0%	93.6%	0.967
Random Forest — test	97.3%	100.0%	96.4%	0.981
Gradient Boosting — labelling	99.2%	100.0%	98.7%	0.994
Gradient Boosting — test	100.0%	100.0%	100.0%	1.000

Tab. 5 Performance of tested models

Gradient Boosting is the best model, misclassifying a single case on the labelling set and classifying every test case correctly (F1 = 1.0). Logistic Regression is the weakest of the four. Notably, the Random Forest reproduces the heuristic's results exactly — the same cases are misclassified — indicating that its decision trees largely replicate the

logical heuristic rule. All models except Logistic Regression achieve 100% precision, consistent with the precision-first design goal.

Fig. 1 shows the main decision tree of the model: we can see that the features it takes into account are the same as in the heuristic (eval_city and eval_occupation), although with a slightly more sophisticated logic

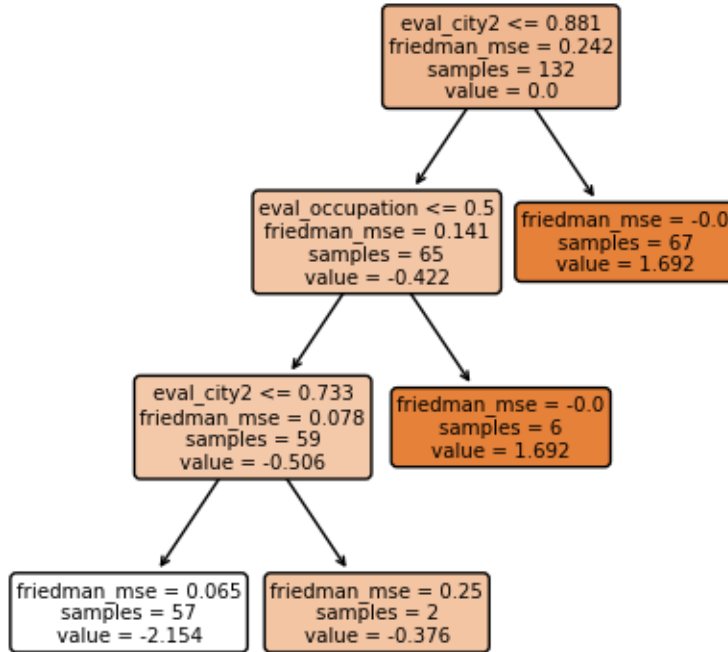


Fig. 1 XGBoost's main decision tree

6.1 Error analysis

The single case Gradient Boosting misclassifies is **Ermenegildo Gasperoni** (San Marino): the Wikidata place-of-birth field is empty — most likely because San Marino is a country with a single city — so the city metric cannot fire, and the occupation fields (bracciante vs. politico) disagree while not being incompatible. Encoding the general knowledge that a single-city country needs no municipality appears to lie beyond any model of reasonable complexity.

The four additional cases misclassified by the heuristic and Random Forest (but correctly handled by Gradient Boosting) share a common cause: in three of them the CPC city field is recorded at a coarser level (province or country) than Wikidata's (municipality), so the city metric fails; one more is the single-city-country situation again. It is notable that Gradient Boosting's more sophisticated tree logic recovers all of these.

6.2 Summary

Across the labelling and test sets, the records of **8,981 persons (5.89% of the CPC)** were processed, of whom 128 (1.43%) could be linked to a Wikidata entry — confirming the extreme long-tail nature of the task. Several automatic labelling models were developed and tested; the best (Gradient Boosting) achieves what we consider very strong performance, with precision close to 100% and only 0.48% false negatives. Finally, an automated pipeline was defined that acquires the required data and performs the linking automatically.